B1 Linear relations and equations: solutions

← Module overview · Problems

Try every problem seriously before reading its solution. When your answer differs, find the exact line where your working first went wrong.

Practice

1. (a)

x	-2	-1	0	1	2	3
y	-11	-8	-5	-2	1	4

(b) Slope $a = 3$ (each step of 1 in x adds 3 to y); start value $b = -5$.

(c) Substitute $x = 4$: $y = 3 \cdot 4 - 5 = 7$. That matches, so $(4, 7)$ lies on the graph.

2. (a) $a = \dfrac{17 - 5}{6 - 2} = \dfrac{12}{4} = 3$.

(b) $y = 3x + b$ through $(2, 5)$: $5 = 6 + b$, so $b = -1$. Answer: $y = 3x - 1$. Check with $(6, 17)$: $18 - 1 = 17$.

3. $a = \dfrac{-12 - 4}{5 - (-3)} = \dfrac{-16}{8} = -2$. Through $(-3, 4)$: $4 = -2 \cdot (-3) + b = 6 + b$, so $b = -2$.

Answer: $y = -2x - 2$. Check: $-2 \cdot (-3) - 2 = 4$ and $-2 \cdot 5 - 2 = -12$. Both correct.

4. $-1 = \frac{1}{2} \cdot 4 + b = 2 + b$, so $b = -3$. Answer: $y = \frac{1}{2}x - 3$.

5. Parallel means the same slope, -3. Through $(2, 4)$: $4 = -3 \cdot 2 + b = -6 + b$, so $b = 10$. Answer: $y = -3x + 10$.

6. (a)

$$\begin{aligned} 5x - 7 &= 2x + 8 \\ 3x - 7 &= 8 \\ 3x &= 15 \\ x &= 5 \end{aligned}$$

Check: $25 - 7 = 18$ and $10 + 8 = 18$.

(b)

$$\begin{aligned} 6x - 3 - 4x - 8 &= 7 \\ 2x - 11 &= 7 \\ 2x &= 18 \\ x &= 9 \end{aligned}$$

Check: $3 \cdot 17 - 4 \cdot 11 = 51 - 44 = 7$.

(c) Multiply both sides by 12:

$$\begin{aligned} 4(x + 1) - 3(x - 2) &= 24 \\ 4x + 4 - 3x + 6 &= 24 \\ x + 10 &= 24 \\ x &= 14 \end{aligned}$$

Check: $\frac{15}{3} - \frac{12}{4} = 5 - 3 = 2$. Note that $-3(x - 2) = -3x + 6$.

(d) Multiply both sides by 10: $4x + 12 = x - 6$, so $3x = -18$, so $x = -6$. Check: $-2.4 + 1.2 = -1.2$ and $-0.6 - 0.6 = -1.2$.

7. (a) $2x + 6 = 2x + 5$; subtracting $2x$ gives $6 = 5$, which is false for every x. **No solution.** (The lines $y = 2x + 6$ and $y = 2x + 5$ are parallel.)

(b) $3x - 6 + x = 4x - 6$, so $4x - 6 = 4x - 6$, which becomes $0 = 0$. True for every x: **every number is a solution**.

8. (a) $3x < 15$, so $x < 5$.

(b) $-2x \geq 8$. Divide by -2 and flip: $x \leq -4$. Check $x = -5$: $5 + 10 = 15 \geq 13$, true.

(c) Multiply both sides by -3 and flip: $x - 1 < -6$, so $x < -5$. Check $x = -8$: $\frac{-9}{-3} = 3 > 2$, true. Check $x = 0$: $\frac{-1}{-3} = \frac{1}{3} > 2$, false. Good.

(d) $2x - 6 - 5x \leq x + 10$, so $-3x - 6 \leq x + 10$, so $-4x \leq 16$. Divide by -4 and flip: $x \geq -4$. (Or: add $3x$ and subtract 10 to get $-16 \leq 4x$, so $-4 \leq x$, without any flipping.)

9. Substitute $y = 2x - 3$: $3x + 2(2x - 3) = 15$, so $7x - 6 = 15$, so $x = 3$. Then $y = 2 \cdot 3 - 3 = 3$.

Solution: $(x, y) = (3, 3)$. Check: $9 + 6 = 15$.

10. Multiply the second equation by 2: $10x - 4y = 68$. Add it to the first: $13x = 78$, so $x = 6$. Then $18 + 4y = 10$, so $4y = -8$, so $y = -2$.

Solution: $(6, -2)$. Check in the second equation: $30 + 4 = 34$.

11. Table of values:

x	0	1	2	3
$y = x + 1$	1	2	3	4
$y = -2x + 7$	7	5	3	1

The lines cross at $(2, 3)$. Calculation: $x + 1 = -2x + 7$, so $3x = 6$, so $x = 2$ and $y = 3$.

12. Write each equation as $y = \ldots$: $2x - 3y = 6$ becomes $y = \frac{2}{3}x - 2$.

(a) $4x - 6y = 12$ becomes $y = \frac{2}{3}x - 2$: the same line. **Infinitely many solutions**: every pair $(x, \frac{2}{3}x - 2)$, for example $(0, -2)$, $(3, 0)$, $(6, 2)$.

(b) $4x - 6y = 10$ becomes $y = \frac{2}{3}x - \frac{5}{3}$: same slope, different start value, so parallel. **No solution.** Algebraically: twice the first equation is $4x - 6y = 12$; subtracting the second gives $0 = 2$.

(c) $3x + 2y = 9$ becomes $y = -\frac{3}{2}x + \frac{9}{2}$: different slope, so **exactly one solution**. (The slopes $\frac{2}{3}$ and $-\frac{3}{2}$ even multiply to -1: the lines are perpendicular.) Elimination: twice the first gives $4x - 6y = 12$, three times the second gives $9x + 6y = 27$; adding gives $13x = 39$, so $x = 3$, then $6 - 3y = 6$, so $y = 0$. Solution: $(3, 0)$.

13. The perpendicular slope is $-\dfrac{1}{2/3} = -\dfrac{3}{2}$ (check: $\frac{2}{3} \cdot -\frac{3}{2} = -1$). Through $(4, 1)$: $1 = -\frac{3}{2} \cdot 4 + b = -6 + b$, so $b = 7$.

Answer: $y = -\frac{3}{2}x + 7$.

School-exam level

14. (a) $K_A = 15 + 0.10t$ and $K_B = 5 + 0.25t$ (and $K_C = 30$).

(b)

minutes	plan A	plan B	plan C
40	€19.00	€15.00	€30.00
120	€27.00	€35.00	€30.00

(c) $15 + 0.10t < 5 + 0.25t$ gives $10 < 0.15t$, so $t > \dfrac{10}{0.15} = 66.67$ (rounded). Plan A is cheaper than plan B **from 67 minutes**. (Check: at 66 minutes A costs €21.60 and B €21.50; at 67 minutes A costs €21.70 and B €21.75.)

(d) A must be cheaper than B, so $t \geq 67$, and cheaper than C: $15 + 0.10t < 30$ gives $t < 150$. So plan A is the cheapest for **67 up to and including 149 minutes**. At exactly 150 minutes A and C both cost €30.

15. (a) The extra 5 km cost $32.60 - 22.10 = 10.50$ euros, so the price per km is $10.50/5 = 2.10$ euros. Start fee: $22.10 - 8 \cdot 2.10 = 22.10 - 16.80 = 5.30$ euros. So $K = 2.10d + 5.30$.

(b) $5.30 + 2.10d \leq 50$ gives $2.10d \leq 44.70$, so $d \leq 21.29$ (rounded). At most **21 km**. (Check: 21 km costs €49.40 and 22 km costs €51.50.)

(c) $3.50 + 2.30d < 5.30 + 2.10d$ gives $0.20d < 1.80$, so $d < 9$. The competitor is cheaper for **rides shorter than 9 km**; at exactly 9 km both cost €24.20.

16. (a) Slope $= \dfrac{212 - 32}{100 - 0} = 1.8$, and at $C = 0$ we have $F = 32$, so the start value is 32. So $F = 1.8C + 32$.

(b) $F = 1.8 \cdot 37 + 32 = 66.6 + 32 = 98.6\,°\text{F}$.

(c) Solve $C = 1.8C + 32$: $-0.8C = 32$, so $C = -40$. So $-40\,°\text{C} = -40\,°\text{F}$.

(d) $C = K - 273.15$, so $F = 1.8(K - 273.15) + 32 = 1.8K - 491.67 + 32 = 1.8K - 459.67$.

17. (a) Revenue $2.30n$, costs $120 + 0.80n$, so $W = 2.30n - 120 - 0.80n = 1.5n - 120$.

(b) $1.5n - 120 = 0$ gives $n = 80$. The break-even point is **80 cupcakes** (at 80 they make exactly €0 profit).

(c) $1.5n - 120 \geq 200$ gives $1.5n \geq 320$, so $n \geq 213.33$ (rounded). They must sell **at least 214 cupcakes**. (213 gives €199.50, 214 gives €201.00.)

18. (a) $L_A = 30 - 2t$ and $L_B = 22 - 1.2t$.

(b) $30 - 2t = 22 - 1.2t$ gives $8 = 0.8t$, so $t = 10$. After **10 hours** both are $30 - 20 = 10$ cm long.

(c) $L_A = 0$ at $t = 15$; $L_B = 0$ at $t = \dfrac{22}{1.2} = 18\frac{1}{3}$ hours $= 18$ h 20 min. Candle **A burns out first**, and B burns out **3 hours 20 minutes** later.

19. (a) $a = \dfrac{4 - 1}{4 - (-2)} = \dfrac{3}{6} = \dfrac{1}{2}$. Through A: $1 = \frac{1}{2} \cdot (-2) + b$, so $b = 2$. Line AB: $y = \frac{1}{2}x + 2$.

(b) Slope $\frac{1}{2}$ through $C(1, -5)$: $-5 = \frac{1}{2} + b$, so $b = -5\frac{1}{2}$. Answer: $y = \frac{1}{2}x - 5\frac{1}{2}$.

(c) Slope -2 (because $\frac{1}{2} \cdot -2 = -1$) through C: $-5 = -2 + b$, so $b = -3$. Answer: $y = -2x - 3$.

(d) $\frac{1}{2}x + 2 = -2x - 3$ gives $\frac{5}{2}x = -5$, so $x = -2$, and $y = 1$. The intersection point is $(-2, 1)$, which is exactly point A. So the line through C perpendicular to AB passes through A: $AC \perp AB$, and **triangle ABC has a right angle at A.**

20. Let v be the number of adult tickets and k the number of child tickets.

$$\begin{cases} v + k = 85 \\ 12v + 7.5k = 858 \end{cases}$$

Substitute $k = 85 - v$: $12v + 7.5(85 - v) = 858$, so $12v + 637.5 - 7.5v = 858$, so $4.5v = 220.5$, so $v = 49$. Then $k = 36$.

49 adult tickets and 36 child tickets. Check: $49 \cdot 12 + 36 \cdot 7.50 = 588 + 270 = 858$.

21. After w weeks Sven has $40 + 6.5w$ and Noor has $95 - 4w$ euros.

$40 + 6.5w > 95 - 4w$ gives $10.5w > 55$, so $w > 5.24$ (rounded). From **week 6** Sven has more.

Check: after 5 weeks Sven has €72.50 and Noor €75.00; after 6 weeks Sven has €79.00 and Noor €71.00.

22. (a) $V_A = 500 - 12t$ and $V_B = 80 + 9t$.

(b) $500 - 12t = 80 + 9t$ gives $420 = 21t$, so $t = 20$. After **20 minutes** both contain $500 - 240 = 260$ litres.

(c) $500 - 12t > 2(80 + 9t)$ gives $500 - 12t > 160 + 18t$, so $340 > 30t$, so $t < 11\frac{1}{3}$. Tank A is empty at $t = \frac{500}{12} = 41\frac{2}{3}$, so the formula for V_A is valid during the whole interval found. Answer: from the start up to $11\frac{1}{3}$ minutes, that is **during the first 11 minutes and 20 seconds**.

Challenge

23. Double the first equation and subtract the second:

$$(4x + 2py) - (4x + 6y) = 10 - q \quad \Longrightarrow \quad (2p - 6)\, y = 10 - q.$$

By §6.5 this equation together with the first equation forms a system equivalent to the original one.

- If $2p - 6 \neq 0$, that is $p \neq 3$, then $y = \dfrac{10 - q}{2p - 6}$ and there is exactly one solution.
- If $p = 3$, the equation reads $0 = 10 - q$.

(a) **No solution** when $p = 3$ and $q \neq 10$. (Then both lines have slope $-\frac{2}{3}$ but the second is not twice the first.)

(b) **Infinitely many** when $p = 3$ and $q = 10$: the second equation is then exactly twice the first, so it is the same line $2x + 3y = 5$.

(c) **Exactly one** when $p \neq 3$ (any q). Then

$$y = \frac{10 - q}{2p - 6}, \qquad x = \frac{5 - py}{2} = \frac{pq - 30}{4p - 12}.$$

(Working for x: $5 - py = \dfrac{5(2p - 6) - p(10 - q)}{2p - 6} = \dfrac{pq - 30}{2p - 6}$, then divide by 2.)

24. $OP^2 = 1^2 + a_1^2$, $OQ^2 = 1^2 + a_2^2$ and $PQ^2 = 0^2 + (a_1 - a_2)^2$.

If $a_1 = a_2$ the two lines coincide, are not perpendicular, and $a_1 a_2 = a_1^2 > 0 \neq -1$, so the statement holds. Assume now $a_1 \neq a_2$, so O, P, Q form a triangle (P and Q are different points on the vertical line $x = 1$, and O is not on that line).

By Pythagoras' theorem and its converse, the angle at O is right exactly when $OP^2+OQ^2 = PQ^2$:

$$\begin{aligned}(1+a_1^2)+(1+a_2^2) &= (a_1-a_2)^2\\ 2+a_1^2+a_2^2 &= a_1^2-2a_1a_2+a_2^2\\ 2 &= -2a_1a_2\\ a_1a_2 &= -1.\end{aligned}$$

Each step can be reversed, so: angle POQ is $90°$ $\iff$ $a_1a_2 = -1$. Since angle POQ is the angle between ℓ_1 and ℓ_2, this proves $\ell_1 \perp \ell_2$ $\iff$ $a_1a_2 = -1$. Lines with the same slopes that do not pass through the origin are parallel to these, so the rule holds for all non-vertical, non-horizontal lines.

25. (a) $y \geq 1$ and $x \geq 1$ means $5y \leq 47-3 = 44$, so $y \leq 8$. For each y from 1 to 8, $47-5y$ must be a positive multiple of 3:

y	1	2	3	4	5	6	7	8
$47-5y$	42	37	32	27	22	17	12	7
multiple of 3?	yes	no	no	yes	no	no	yes	no

So the solutions are $(x,y) = (14,1)$, $(9,4)$ and $(4,7)$. We checked every possible y, so there are no others. Notice the pattern: going from one solution to the next, x drops by 5 and y rises by 3, which keeps $3x+5y$ the same ($-15+15=0$).

(b) $8 = 3+5$, $9 = 3+3+3$, $10 = 5+5$. Now every $n \geq 11$ is 3 more than a number $n-3 \geq 8$. Starting from 8, 9 and 10 and adding packs of 3 again and again, we reach 11, 12, 13, then 14, 15, 16, and so on: every whole number $n \geq 8$. (This is a proof by induction.)

Impossible amounts below 8: 3, 5 and 6 are possible (3, 5, $3+3$). For 1, 2 and 4 the only multiples of 5 that fit are 0, and 1, 2, 4 are not multiples of 3. For 7: using 0 packs of 5 needs $7 = 3x$, impossible; using one pack of 5 leaves 2, not a multiple of 3. So the impossible amounts are exactly **1, 2, 4 and 7**.

26. Add the first and third equation: $2x+3y = 8$. Add the second and third: $3x+y = 5$. Now z is gone. From $3x+y=5$: $y = 5-3x$. Substitute: $2x+3(5-3x) = 8$, so $-7x = -7$, so $x = 1$, $y = 2$. From the first equation, $z = 6-1-2 = 3$.

Solution: $(x,y,z) = (1,2,3)$. Check: $1+2+3 = 6$, $2-2+3 = 3$, $1+4-3 = 2$.

27. Write E_1 as $L_1 = R_1$ and E_2 as $L_2 = R_2$.

(a) If (x,y) satisfies E_1 and E_2, then adding k times the true statement $L_1 = R_1$ to the true statement $L_2 = R_2$ gives the true statement $L_2 + kL_1 = R_2 + kR_1$. Conversely, if (x,y) satisfies E_1 and $L_2 + kL_1 = R_2 + kR_1$, then subtracting k times $L_1 = R_1$ gives $L_2 = R_2$. So the two systems have exactly the same solutions.

(b) If $L_1 = R_1$ then $mL_1 = mR_1$. Conversely, if $mL_1 = mR_1$ and $m \neq 0$, divide by m to get $L_1 = R_1$. The second equation is unchanged. So the systems have the same solutions.

(c) Take E_1: $x+y = 2$ and E_2: $x-y = 0$. This system has exactly one solution, $(1,1)$. Multiplying E_1 by $m = 0$ gives $0 = 0$, and the system $\{0=0,\ x-y=0\}$ is solved by every pair with $x = y$, for example $(5,5)$. Solutions were gained, because the step "divide by m" cannot be done when $m = 0$.

28. (a) Rewrite: $y = p(x+2)-3$, so $y+3 = p(x+2)$. For $x = -2$ the right-hand side is 0 whatever p is, so $y = -3$. So every line ℓ_p passes through $(-2,-3)$. (It is the only common point, because for example ℓ_0: $y = -3$ and ℓ_1: $y = x-1$ meet only there.)

(b) If $p = 0$, the line is $y = -3$, which does not meet the x-axis. If $p \neq 0$, the line meets the x-axis where $0 = p(x+2) - 3$, so $x = \dfrac{3}{p} - 2$. We need $0 \leq \dfrac{3}{p} - 2 \leq 4$, that is $2 \leq \dfrac{3}{p} \leq 6$. Since $\dfrac{3}{p} \geq 2 > 0$, p must be positive, so we may multiply by p without flipping: $2p \leq 3 \leq 6p$. So $p \leq \frac{3}{2}$ and $p \geq \frac{1}{2}$.

Answer: $\frac{1}{2} \leq p \leq \frac{3}{2}$.

Geometric check: all lines pass through $(-2, -3)$. The line to $(0, 0)$ has slope $\frac{3}{2}$ and the line to $(4, 0)$ has slope $\frac{3}{6} = \frac{1}{2}$. Lines with slopes in between sweep across the segment.

29. Let the tens digit be t and the units digit e, so the number is $10t + e$ and the swapped number is $10e + t$.

$$\begin{cases} t + e = 11 \\ (10e + t) - (10t + e) = 27 \end{cases}$$

The second equation simplifies to $9e - 9t = 27$, so $e - t = 3$. Adding this to $t + e = 11$ gives $2e = 14$, so $e = 7$ and $t = 4$. The number is **47** (check: $4 + 7 = 11$ and $74 - 47 = 27$). The system has exactly one solution (the lines $t + e = 11$ and $e - t = 3$ have different slopes), so 47 is the only such number.

30. (a) $2x + 1 = -x + 7$ gives $x = 2$, $y = 5$. The third line must pass through $(2, 5)$: $5 = \frac{1}{2} \cdot 2 + k = 1 + k$, so $k = 4$.

(b) The vertices of the triangle are: the intersection point $(2, 5)$; where $y = 2x + 1$ meets the x-axis, $x = -\frac{1}{2}$; where $y = -x + 7$ meets the x-axis, $x = 7$. The base on the x-axis has length $7 - (-\frac{1}{2}) = 7\frac{1}{2}$ and the height is the y-coordinate of the top vertex, 5. Area $= \frac{1}{2} \cdot 7\frac{1}{2} \cdot 5 = \dfrac{75}{4} = 18\frac{3}{4}$.